Atharva Chandak

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WORK EXPERIENCE

EBAY **SEPT 2025 – PRESENT**

ML Engineer 2 | Python, Pytorch, HuggingFace, vLLM, LangChain, Triton *Bengaluru, India*

- Built high-precision **Multi-Modal models** to match seller listings to catalog products for recommendations, **boosting GMV by $20 Mil**.
- Researching and **finetuning SOTA Vision LLMs** including Qwen3, Gemma4, etc. for **fine-grained image extraction** in Collectibles segment.
- Optimised and deployed LLMs **improving TPS by 30%** achieving better throughput and GPU utilisation and reducing long term costs.
- Improved the catalog matching **precision by 2.5%**, **coverage by 4%** and **recall by 2%** across multiple categories improving the end-to-end UX.
- Prototyped a **RAG-based** buyer experience assistant for **hyper-personalised** recommendations and **agentic assistance** to users using memory.

WELLS FARGO **JUL 2023 – SEPT 2025**

SWE | Python, Pytorch, Java, JavaScript, React, Spring, Jenkins, MongoDB *Bengaluru, India*

- Built a **RAG-based LLM assistant** from prototype to production for merchant workflows, reducing loan application time by **upto 30%**.
- Engineered credit-card services, increasing transaction throughput processing **1M+ transactions/month** across 18K+ merchants.
- Developed scalable POS EMI microservices and **merchant analytics systems** supporting high-throughput workflows.

MILA / ROBOTICS AND EMBODIED AI LAB, UNIVERSITY OF MONTREAL **JAN 2023 – JUN 2023**

Research Assistant | Python, Pytorch, Hugging Face *Montreal, Canada*

- Researched **Continual Object Detection** in autonomous driving for my undergrad thesis, cutting catastrophic forgetting in OD by **73%**.
- Created on-device incremental OD methods for robotics demonstrating **mAP50 drop** of only **3.99** for transformer models vs CNNs' 14.84.
- Achieved a domain incremental **avg mAP50 of 68.63** and **avg mAP of 43.89** on the CLAD-D dataset marking **15% gain** over baselines.

WELLS FARGO **MAY 2022 – JUL 2022**

Intern | JavaScript, Python, React, Flask, MongoDB *Hyderabad, India (Remote)*

- Designed and implemented an **AI-first voice chatbot** for home lending, reducing query resolution time by **up to 68%**.
- Integrated **speech-to-text**(STT) and **TTS** APIs for voice interaction, and built an action server for NLP-driven processing & response curation.
- Composed the **NLP pipeline** for intent and entity extraction and achieved an **F1 scores** of **93.4%** and **83.5%**.

AIRLAB, CARNEGIE MELLON UNIVERSITY **AUG 2021 – DEC 2021**

Intern | Python, Pytorch | PAPER LINK *Pittsburgh, USA (Remote)*

- Implemented **the vision pipeline** for TartanAviation, generating **3.1M** high-resolution images across 550 terminal airspace sequences.
- Engineered calibration and **auto-labeling frameworks** for precise aircraft detection, producing **700k+** high-fidelity bounding box annotations.

EDUCATION

BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE PILANI (BITS PILANI)

Bachelor of Engineering in Computer Science

Coursework : Data Structures & Algorithms (DSA), Object Oriented Programming (OOP) design patterns, Database Systems, Computer Networks, Cloud Computing, Operating Systems, Probability & Statistics, Machine Learning, Deep Learning, Computer Vision (CV), Natural Language Processing

PUBLICATIONS AND PROJECTS

ADQUESTA: KNOWLEDGE-GUIDED VQA FRAMEWORK FOR ADVERTISEMENTS | PAPER LINK **WACV, FEB 2025**

- Proposed a novel VQA framework achieving **pattern-matching** accuracy of **58.58%** on ads, outperforming 10x larger models.
- Introduced **first** ad-specific **VQA dataset** featuring **94k+ QA pairs** & **17k images**, establishing a new benchmark for ad comprehension research.
- Integrated **RAG** and **LLaVA**-based implicit knowledge, leading to **2X increase** in QA accuracy compared to previous state-of-the-art approaches.

LIFTNET: ACTION RECOGNITION ACROSS VARIED LIGHTING ENVIRONMENTS | PAPER LINK **MAY 2022 – DEC 2022**

- Architected LIFT-Net, a **dual-pathway transformer**, exceeding SOTA by **2.20%** (HMDB51) and **0.12%** (UCF101) for action recognition.
- Enhanced **illumination robustness** by efficient spatiotemporal modeling, achieving **95.21%** on ARID (+**1.14%**) and **2.65%** gain on InfAR.

TECHNICAL SKILLS

ML / AI : Multimodal Learning, Vision Large Language Models/LLMs, CV, NLP, VQA, RAG, RL, AI Agents, OD, Retrieval & Re-ranking, Continual Learning
Frameworks : Pytorch, Hugging Face, LangChain, NumPy, pandas, OpenCV, FastAI, scikit-learn, Pytorch-lightning
Languages : Python, JavaScript, TypeScript, Java, C++
Systems : Triton, Docker, Linux, AWS /Azure Cloud, Kafka, MongoDB, Jenkins (CI/CD), Git, ROS

LEADERSHIP / VOLUNTEERING

BITS-ACM Core Team Managed a software development hackathon (HackBITSPilani); Contributed and managed org's open source projects.
Senior Member, ShARE BITS Pilani Interned at Catalytic Corps, developing a framework to analyze and solve Indian MSME sector challenges.
Teaching Assistant Orchestrated ML/DL course assignments and conducted workshops on programming neural nets and foundational AI concepts.
NSS Volunteer Taught computer literacy and organized social impact events like blood donation camps, youth conferences, etc. in Pilani.